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Article

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Posted Date: July 10th, 2024

DOI: <https://doi.org/10.21203/rs.3.rs-4601143/v1>

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Additional Declarations: No competing interests reported.

Enhanced Raccoon Optimization Algorithm for PMSM Electrical Parameter Identification

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ABSTRACT

This article proposed an improved algorithm for the parameter identification of permanent magnet synchronous motors (PMSMs). An enhanced raccoon optimization algorithm (EROA) was formed by combining the raccoon optimization algorithm (ROA) with the adaptive exploration radius strategy, raccoon-washing-food-inspired strategy, and escaping-predator strategy. First, using some of the functions in IEEE CEC2015, the EROA solution has a large improvement in convergence speed and solution accuracy compared with other algorithms. Second, the EROA solution is more stable under the same conditions as demonstrated by MATLAB parameter identification simulation. Finally, EROA applies to motor parameter identification through motor control experiments.

Keywords PMSM, parameter identification, ROA improvement

Introduction

Permanent magnet synchronous motors (PMSMs) are widely used in industry due to their high efficiency and small size.¹⁻³ However, realizing the high-performance control for PMSMs requires accurate motor parameters. For example, model predictive control^{4,5} and speed sensorless control^{6,7} require accurate motor parameters as a basis. The PMSM parameter identification is categorized into online identification⁸⁻¹⁰ and offline identification¹¹⁻¹³. Online identification is often combined with motor control methods^{14,15} to update the motor parameters. Offline identification is used to initialize parameters before motor startup¹⁶, which is widely used in the industry.

The main online identification methods are recursive least squares (RLS), model reference adaptive (MRAS), extended Kalman filtering, and high-frequency signal injection. RLS has good mathematical-statistical properties and a wide range of adaptability. In article¹⁷, inductance, flux linkage, and resistance parameters are recognized by a recursive total least-square excitatory and inhibitory learning to update the model parameters of the velocity-free sensor. However, RLS must measure the convergence speed and solution accuracy of the algorithm. In article¹⁸, inductance, resistance, and flux linkages are recognized by MRAS. However, the adaptive rate design process of MRAS is more complex. In article¹⁹, the system state and two required motor parameters are estimated using a dual extended Kalman filter. In article²⁰, the motor parameters are obtained by injecting high-frequency sinusoidal voltages into the stationary coordinate system ($\alpha\beta$ axis) to update the model parameters in the position observer.

Offline identification methods include finite element analysis (FEA)²¹, signal injection²², and artificial intelligence algorithms²³. FEA can provide high-precision parameters that can well reflect the nonlinear characteristics of the motor, but FEA is commonly used in the pre-design stage of the motor. In article²⁴, the effectiveness of its parameter identification scheme is verified by FEA. In article¹⁶, the amplitude-auto-adjusting signal injection (ASI) method is proposed to recognize the inductance and stator resistance in different states offline. Intelligent algorithms have rapidly developed, and many types of intelligent algorithms have emerged²⁵. Depending on the source of inspiration, they can be categorized as swarm-based, evolution-based, physics-based, game-rule-based, and human social-relationship-based²⁶. Artificial intelligence algorithms have outstanding ability to solve high-dimensional functions, and scholars have applied them to the solution of motor parameters²⁷. Commonly, an improved particle swarm algorithm (PSO) performs the parameter identification. In article²⁸, a modified PSO is used to solve the motor parameter identification objective function. In article²⁹, the particle motion states are improved to form an

interactive and dynamically learned PSO to solve for motor parameters.

ROA³⁰ has a better solution at lower population sizes. In article³¹, ROA reduces the scheduling time compared to other algorithms. In article³², an algorithm based on ROA is implemented for precise localization to improve the accuracy of the range measurements. In article³³, ROA is used to significantly improve job scheduling. By improving the ROA, motor parameters can be identified in this article.

In this article, the signal injection method combined with intelligent algorithms is used for PMSM parameter identification. Since the direct application of ROA cannot solve the problem well, this article improves ROA using the adaptive exploration radius strategy, raccoon-washing-food-inspired strategy, and escaping-predator strategy. To validate the effectiveness of EROA, the more classical algorithms of ROA, PSO³⁴, E_WOA³⁵, and GA³⁶ were selected for comparison using some of the IEEE CEC 2015 test sets. EROA has a large advantage in convergence speed and solution accuracy. Parameter identification objective functions are established in simulation and motor experiments for comparison. EROA can more quickly find the motor parameters.

Parameter Identification of PMSM

The parameter identification method in this article is based on the voltage equations in the steady state of the PMSM. To facilitate the analysis, the motor voltage equation (1), which ignores the effect of core saturation and losses of PMSM, is considered in the dq rotating coordinate system.

$$\begin{cases} u_d = R_s i_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q \\ u_q = R_s i_q + L_q \frac{di_q}{dt} - \omega_e L_d i_d + \psi_f \omega_e \end{cases} \quad (1)$$

where u_d and u_q are the dq-axis voltages; i_d and i_q are the dq-axis currents, ω_e are the electromechanical angular velocities; R_s is the stator resistance; L_d and L_q are the dq-axis inductances; ψ_f is the permanent flux linkage; ω_e is the electrical angular velocity.

Through PMSM vector control, the motor steadily operates at a certain speed. At this point, the dq-axis currents of the PMSM remain stable in a very small range. It may be assumed that $\frac{di_d}{dt} = \frac{di_q}{dt} = 0$. From Equation (1), we obtain Equation (2).

$$\begin{cases} u_d = R_s i_d - \omega_e L_q i_q \\ u_q = R_s i_q - \omega_e L_d i_d + \psi_f \omega_e \end{cases} \quad (2)$$

Injecting different currents into the d-axis increases the number of equations. The system of full-rank voltage equations is shown in (3), which can be used as a reference model for the motor.

$$\begin{cases} u_{d0} = -\omega_{e0} L_q i_{q0} \\ u_{q0} = R_s i_{q0} + \psi_f \omega_{e0} \\ u_{d1} = R_s i_{d1} - \omega_{e1} L_q i_{q1} \\ u_{q1} = R_s i_{q1} - \omega_{e1} L_d i_{d1} + \psi_f \omega_{e1} \end{cases} \quad (3)$$

where $id = 0$ defines the formula subscript as 0, and $id = -1$ defines the formula subscript as 1. The adjustable model (4) is constituted by the PMSM run data through the reference model.

$$\begin{cases} \hat{u}_{d0} = -\omega_{e0} \hat{L}_q i_{q0} \\ \hat{u}_{q0} = \hat{R}_s i_{q0} + \hat{\psi}_f \omega_{e0} \\ \hat{u}_{d1} = \hat{R}_s i_{d1} - \omega_{e1} \hat{L}_q i_{q1} \\ \hat{u}_{q1} = \hat{R}_s i_{q1} - \omega_{e1} \hat{L}_d i_{d1} + \hat{\psi}_f \omega_{e1} \end{cases} \quad (4)$$

Using the sum of squares of the differences between reference model and adjustable model, we obtain (5) as a fitness function for parameter identification.

$$\begin{cases} f_1(L_q) = [u_{d0}(k) - \hat{u}_{d0}(k)]^2 \\ f_2(R_s, \psi_f) = [u_{q0}(k) - \hat{u}_{q0}(k)]^2 \\ f_3(R_s, L_q) = [u_{d1}(k) - \hat{u}_{d1}(k)]^2 \\ f_4(R_s, L_d, \psi_f) = [u_{q1}(k) - \hat{u}_{q1}(k)]^2 \end{cases} \quad (5)$$

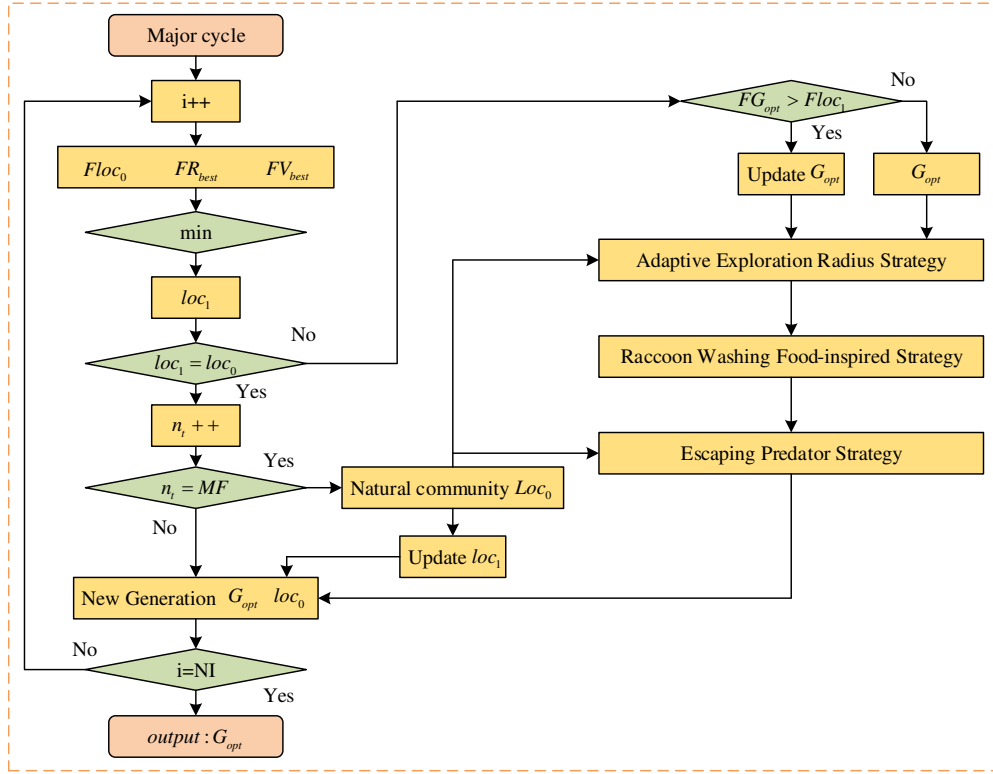


Figure 1. EROA algorithm main loop.

When suitable motor parameters are encountered such that the sum of the four functions is 0, the PMSM parameters in (6) are found.

$$Fun(p) = \sum_{i=1}^4 f_i(p), \quad p = [Rs, L_d, L_q, \psi_f] \quad (6)$$

$Fun(p)$ is the fitness of the parameter vector p . In this paper, this function is solved by improving the raccoon optimization algorithm, and the final parameter vector p can be obtained when the function converges to a certain range.

Enhanced ROA

Characteristics of ROA

The ROA algorithm is mainly characterized by generating two sets of scenarios R_{best0} and V_{best0} in two ranges around the initial location loc_0 : reachable zone radius (RZR) and visible zone radius (VZR). We compare the fitness values of R_{best0} , V_{best0} , and loc_0 to obtain their best solution (7).

$$Fun(loc_1) = \min[Fun(loc_0), Fun(R_{best}), Fun(V_{best})] \quad (7)$$

By comparing the fitness value of the new generation loc_1 with G_{opt} , the algorithm determines whether to update the globally optimal solution G_{opt} . If no good solution is selected around loc_0 , we infer that $loc_1 = loc_0$, which represents no movement in the raccoon location and requires a cumulative migration factor (MF). When MF reaches a certain number of times, the loc_1 position is forced to update. If a certain number of times is not reached, we generate a different population. We perform the iteration NI times and finally output G_{opt} . To quickly find the optimal solution, it is necessary to rationalize the migration scheme and new position after the migration.

To improve the algorithm accuracy, the ROA is improved using the adaptive exploration radius strategy, raccoon-washing-food-inspired strategy, and escaping-predator strategy to form EROA. Figure 1 shows the overall flow of EROA. In Figure 1, i is the current number of iterations, n_t is the current number of times $loc_i = loc_0$, Loc_0 is the raccoon natural community, and $Floc_0$, $Floc_1$, FR_{best} , and FV_{best} are the values of loc_0 , loc_1 , R_{best} , and V_{best} for the fitness values, respectively.

Adaptive Exploration Radius Strategy

The algorithm uses the adaptive exploration radius strategy to facilitate the solution because the initial radius of exploration in the algorithm is a human-set initial value, but each dimension is almost unpredictable for the problem that requires a solution. To improve the convergence accuracy of the algorithm, one approach is to slowly converge the exploration radius during the iteration of the algorithm. The second approach is to explore different positions so that the algorithm can be automatically adjusted to the appropriate exploration radius.

The strategy to decrease the exploration radius with the number of iterations is expressed as (8).

$$\begin{cases} RZR = (X_{max} - X_{min}) \times k & \text{if}(i < N_i) \\ RZR = (X_{max} - X_{min}) \times k \div i & \text{if}(i \geq N_i) \end{cases} \quad (8)$$

where X_{min} and X_{max} constitute the exploration range, and k controls the range of RZR initial values. To adapt the radius of exploration so that the algorithm can automatically learn, define (9) as follow:

$$R_r = G_{opt} - Loc(round(Nop), 1) \quad (9)$$

where Loc_0 is the raccoon community, Nop is the set population density, R_r is the adaptive radius updated with iterations, and $round$ is defined to generate random positive integers. To ensure that R_r has self-learning properties, we set (10).

$$FLoc_r = Fun(G_{opt} + rand \times R_r) \quad (10)$$

where $FLoc_r$ is the evaluated value of another raccoon that arrives the new location under the influence of G_{opt} ; $rand$ denotes a 0-1 uniform distribution. Adaptive updating of RZR is performed during the iteration of the algorithm as shown in (11).

$$RZR = R_r, \text{ if}(FLoc_r < FG_{opt}) \quad (11)$$

If G_{opt} is updated, the adaptive radii RZR and VZR are updated with G_{opt} .

Raccoon-Washing-Food-inspired Strategy

The raccoons must find a nearby water source to wash their food. From this perspective, the behavior of raccoons searching for wetlands after capturing food is modeled. Assume that the current optimal raccoon is the earliest to find food and arrives at the water source to wash the food, and the other raccoons come relatively late. At this point, the cluster of raccoons at the water edge can be represented as a partial cluster near the optimal raccoon, which produces a better solution by combining the optimal raccoon with the later ones in conjunction with one another. A mathematical model of the process in which the optimal raccoon and other raccoons head out to a nearby water source is established as follows. The small cluster $gRZR$ (12) represents the concentration of raccoons near a given water source.

$$gRZP = \begin{bmatrix} gRZP_1 \\ \vdots \\ gRZP_i \\ \vdots \\ gRZP_m \end{bmatrix}_{m \times 1} \quad (12)$$

where the new location of each raccoon is updated according to (13).

$$gRZP_i = G_{opt} + (rand - 0.5) \times (G_{opt} - Loc(s, :)) \quad (13)$$

$Loc(s, :)$ is the location of another raccoon with the same objective. The smallest $FgRZP_{min}$ (14) is selected in $gRZR$ (12).

$$FgRZP_{min} = \min(Fun(gRZP_i)) \quad (14)$$

The size of the fitness of $gRZR_{min}$ to G_{opt} determines whether G_{opt} is updated, as shown in equation (15).

$$G_{opt} = gRZP_{min}, \text{ if}(Fun(gRZP_{min})) < FG_{opt} \quad (15)$$

This produces better results and improves the algorithm solution accuracy.

Escaping-Predator Strategy

A raccoon ducking under attack often creates new possibilities. The raccoon escape strategy is modeled by assuming that a raccoon looks for a direction to quickly escape after being attacked. During this behavior, the ability of the raccoon to arrive at a new safe location implies that the algorithm finds a new target. Failure to do so means that the raccoon fails to escape and a better target is not found. Mri is the new location of the raccoon after it has escaped, and its update process can be tabulated as shown in (16).

$$Mri = Loc_0(xi, :) + (MXmin + (rand + 0.5) \times (MXmax - MXmin)), MXmax = \frac{X_{max}}{i}, MXmin = \frac{X_{min}}{i} \quad (16)$$

$MXmin$ and $MXmax$ are the current maximum range of escape of the raccoon, which continuously changes as the algorithm iterates; xi is a random integer, and $0 < xi \leq Nop$. The resulting superior solution is used to update the raccoon colony (17).

$$\begin{cases} Loc_0(xi, :) = Mri \\ FLoc_{xi} = Fun(Mri) \end{cases}, \text{ if } (Fun(Mri) < Fun(Loc_0(xi, :))) \quad (17)$$

which can be compared with G_{opt} (18).

$$\begin{cases} G_{opt} = Mri \\ FG_{opt} = Fun(Mri) \end{cases}, \text{ if } (Fun(Mri) < FG_{opt}) \quad (18)$$

Here, the algorithm can greatly jump out of the local optimum and find better results.

Simulation and Experimental Methods

Algorithm Testing

To validate the EROA effectiveness, the algorithm was tested through some of the IEEE CEC 2015 function test sets. Five algorithms were set to identical conditions with a total population size of 100 and 1000 iterations: ROA, PSO, E_WOA, GA, and EROA. To approach the dimensionality of the problem of parameter identification, we tested the algorithms by selecting their medium fixed-dimensional multi-peaked test functions $F14 - F23$.

Table 1 shows the values of the fixed-dimensional multi-peak test functions $F14-F23$.

Table 1. IEEE CEC2015 $F14 - F23$ test function set.

Function	Dimensionality	Minimum Value
$F14$	2	0.998
$F15$	4	0.0003075
$F16$	2	-1.0316285
$F17$	2	0.398
$F18$	2	3
$F19$	3	-3.86
$F20$	6	-3.32
$F21$	4	-10.153
$F22$	4	-10.403
$F23$	4	-10.536

Figure 2 shows the effect of the test function algorithm iteration. Overall, the EROA algorithm could quickly find the optimal value during the solution process. Because there is chance in the heuristic class algorithm, in $F23$, EROA found the optimal value after 500 generations. Compared with ROA, EROA has a large improvement in solving speed and accuracy. Compared with algorithms such as E_WOA, EROA is leading in $F14 - F22$. Through 10 independent experiments, the average of the algorithm solution results was obtained for comparison, and Table 2 shows the results. As shown in Table 2, there is a significant improvement in EROA compared to ROA. In $F14$, $F16$, $F17$, $F18$ and $F19$, EROA, PSO and E_WOA have identical effects. In $F15$, $F21$ and $F23$, the value sought by EROA is minimized. In $F22$, EROA and PSO have similar effects. Overall, EROA excels compared to other algorithms.

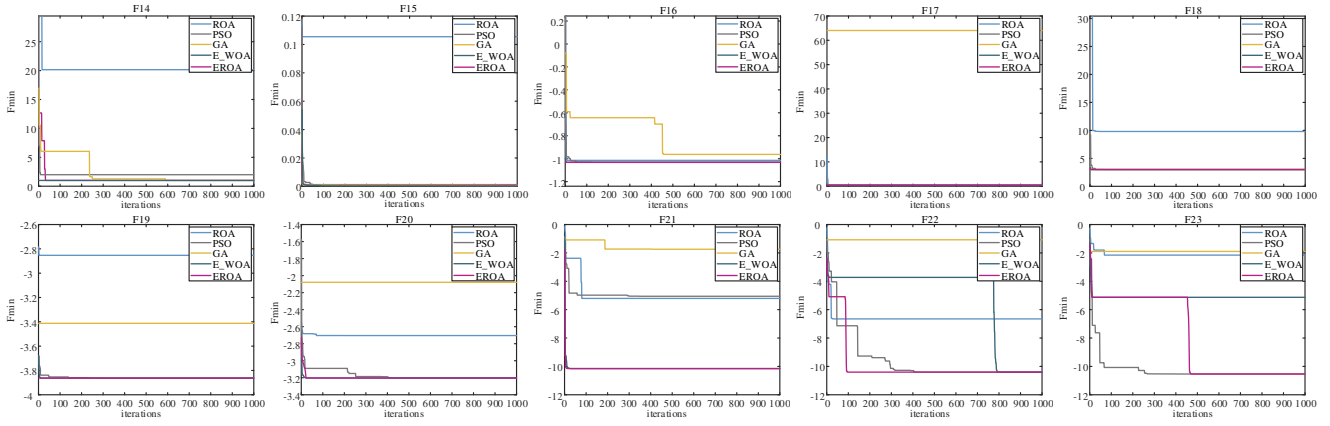


Figure 2. Comparison of EROA and the typical algorithm to test the result graph of 1000 iterations.

Table 2. Comparison of EROA with the typical algorithm to test the average value of 10 iterations.

Function	EROA	ROA	PSO	E_WOA	GA
<i>F14</i>	9.98E-01	1.12E+01	9.98E-01	9.98E-01	1.14E+00
<i>F15</i>	4.91E-04	5.29E-03	6.33E-04	5.01E-04	1.06E-02
<i>F16</i>	-1.03E+00	-9.62E-01	-1.03E+00	-1.03E+00	-1.01E+00
<i>F17</i>	3.98E-01	4.42E-01	3.98E-01	3.98E-01	6.61E+01
<i>F18</i>	3.00E+00	1.20E+01	3.00E+00	3.00E+00	3.00E+00
<i>F19</i>	-3.86E+00	-3.33E+00	-3.86E+00	-3.86E+00	-3.46E+00
<i>F20</i>	-3.29E+00	-1.77E+00	-3.27E+00	-3.30E+00	-2.03E+00
<i>F21</i>	-9.65E+00	-5.25E+00	-8.13E+00	-8.62E+00	-1.67E+00
<i>F22</i>	-9.87E+00	-7.07E+00	-9.87E+00	-7.75E+00	-1.65E+00
<i>F23</i>	-9.45E+00	-7.29E+00	-8.92E+00	-7.83E+00	-1.52E+00

Simulation Analysis

The PMSM vector control simulation model was built in MATLAB. The motor in this experiment was a surface-mounted PMSM. In the process of identifying the motor parameters, let $L_d = L_q = L_s$. Table 3 shows the PMSM parameters. We set the simulation parameters as follows: DC bus voltage: 150 V; motor angular velocity speed: 40 Hz; 0.5 s to join the load. Throughout the control process, a current with $i_d = 0A$ and $i_d = -1A$ was injected into the d -axis. The data during the operation of the motor were saved to the MATLAB workspace, such as i_d, i_q, u_d, u_q , and ω_e . To test the algorithm, the algorithm population size was set to 20, and the number of iterations was set to 200. Seventy data sets were selected for parameter identification. Table 4 shows the average values of the solution effectiveness of the five algorithms such as EROA. From Table 4, the EROA algorithm had the highest solution accuracy with three parameters within 1% error. ROA had large errors in L_s and ψ_f , which could not be satisfied. PSO had a similar solving accuracy to EROA. The L_s error of the E_WOA solution was large, which shows that the EROA solution is more stable than the E_WOA solution under identical conditions.

Experimentation

The motor control platform was the TMDSHVMTRPFCKIT from Texas Instruments (TI). Figure 3 shows the physical motor control platform. The experimental platform was a PMSM control system with TMS320F28335 as the core. The PMSM model number was EMJ-04APB22, and Table 3 shows the main parameters. Motor motion state control was performed using Code

Table 3. PMSM data sheet.

Parameters	Value	Parameters	Value
$R_s(\Omega)$	2.35	Rated Current(A)	2.7
$L_s(H)$	0.0265	Rated Torque(N·m)	1.27
$\psi_f(Wb)$	0.0101	Rated Speed (rpm)	3000
Pole Pairs	4	Rated Power(W)	400

Table 4. Five algorithms for solving statistical tables.

Algorithm	$R_s(\Omega)$	$L_s(H)$	$\psi_f(Wb)$
Real Value	2.35	0.0265	0.0101
EROA	2.3506	0.0264	0.0100
EROA error (%)	0.0256	-0.5472	-0.9527
ROA	2.3753	0.0176	0.0037
ROA error (%)	1.0750	-33.5062	-63.0814
PSO	2.3506	0.0263	0.0100
PSO error (%)	0.0269	-0.6137	-1.0772
E_WOA	2.3555	0.0248	0.0102
E_WOA error (%)	0.2323	-6.3166	0.6497
GA	1.9989	0.2082	0.1258
GA error (%)	-14.9384	685.7044	1145.9741

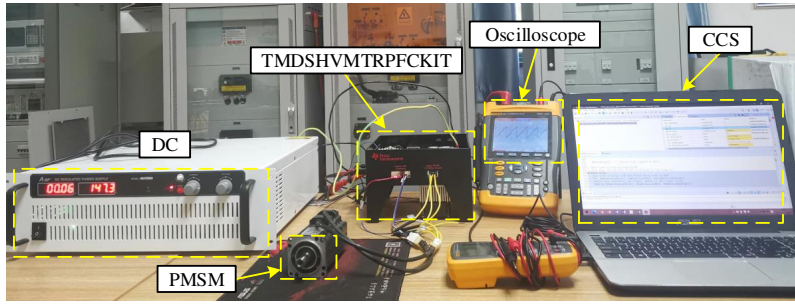


Figure 3. TMDSHVMTRPFCKIT experimental bench.

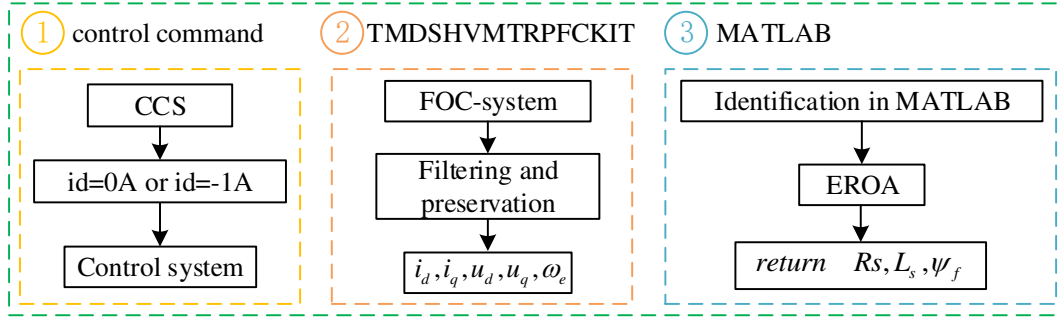


Figure 4. Experimental process.

Composer Studio (CCS). The inverter DC bus voltage was set to 150 V, and the Interrupt Service Routine frequency of 10 kHz was defined. A segment of motor operating data, including the dq -axis voltage, current, and ω_e , was saved while the motor was stabilized. Due to the tight storage resources inside the chip, 70 sets of motor operation data were saved for this experiment. The operating data in the control chip was uploaded to the host computer through CCS, and the motor recognition function was implemented in MATLAB. Figure 4 shows the motor parameter identification process. In this process, a set of experimental data was selected to compare the effects of EROA, ROA, PSO, GA and E_WOA. The algorithm parameters were set to populate 20 and 300 iterations. Figure 5 shows the performance of the five intelligent algorithms.

When EROA was used to solve for the motor parameters, the convergence speed was larger, more accurate and more stable compared to those of the other four medium algorithms. PSO had similar solution accuracy to EROA, but EROA converged more quickly and stabilized at approximately 150 iterations. This algorithm maintained a high accuracy with few iterations and obtained a more algorithm solution. This result is consistent with the simulation. Figure 6 shows the parameter identification results for 70 sets of data.

The EROA identification had a more stable effect than the identification of other algorithms. Table 5 shows the statistical analysis results of 70 sets of data.

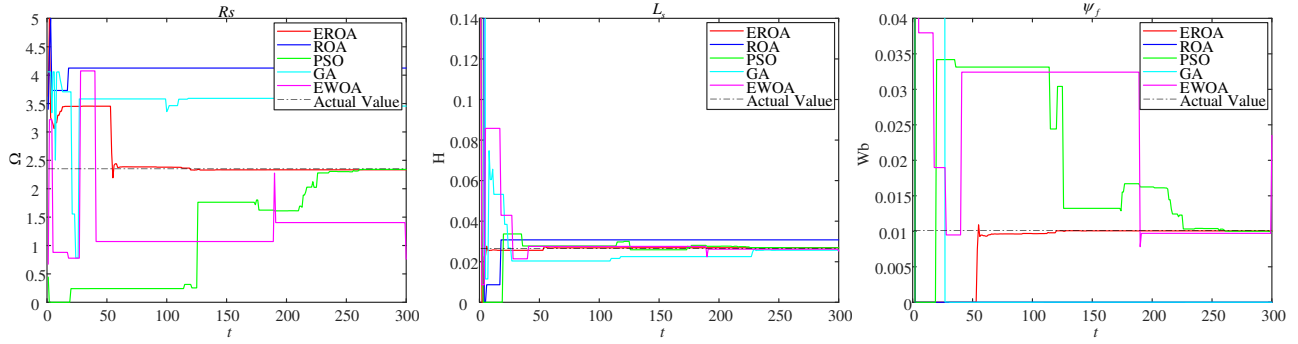


Figure 5. Comparison diagram of the algorithm iteration process.

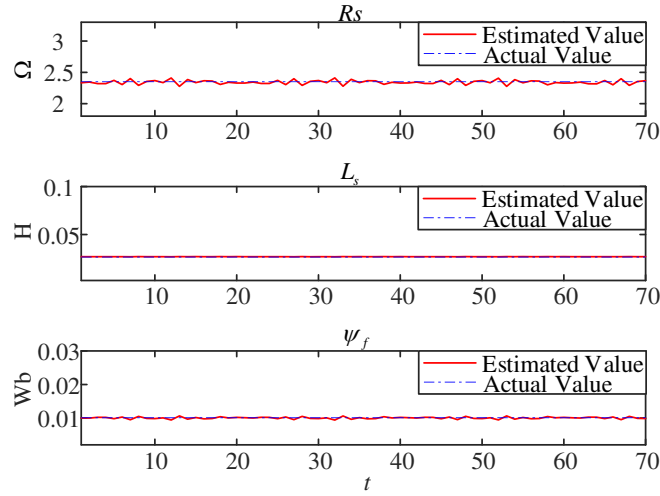


Figure 6. Results of 70 sets of data identification in MATLAB.

As shown in Table 5, the improved raccoon algorithm EROA in this article was used to recognize the motor parameters with average errors of 1% for R_s and ψ_f and 1.6% for L_s .

In this article, the EROA algorithm was applied to the motor control system to identify the motor parameters on the motor control platform TMDSHVMTRPFCKIT. Figure 7 shows the parameter identification process. We controlled the motor motion state and saved a portion of the motor operation data. The EROA solving algorithm was applied in the motor control chip to solve the motor parameters. To ensure the accuracy of the EROA solution, the population was set at 20, and the number of iterations was set at 400. The obtained motor parameters were imported into the computer for review, and Figure 8 shows the results. The data therein were analyzed, and Table 6 shows the results. The parameter identification in the motor control system was performed using EROA. The error of the required parameters was within 2%, and the motor parameters were obtained.

Table 5. Statistical results of the MATLAB identification of 70 sets of data.

EROA	$R_s(\Omega)$	$L_s(H)$	$\psi_f(Wb)$
actual value	2.35	0.0265	0.0101
maximum	2.4105	0.0270	0.0107
minimum	2.2752	0.0268	0.0094
average	2.3425	0.0269	0.0100
average error/%	-0.3193	1.5064	-0.6948

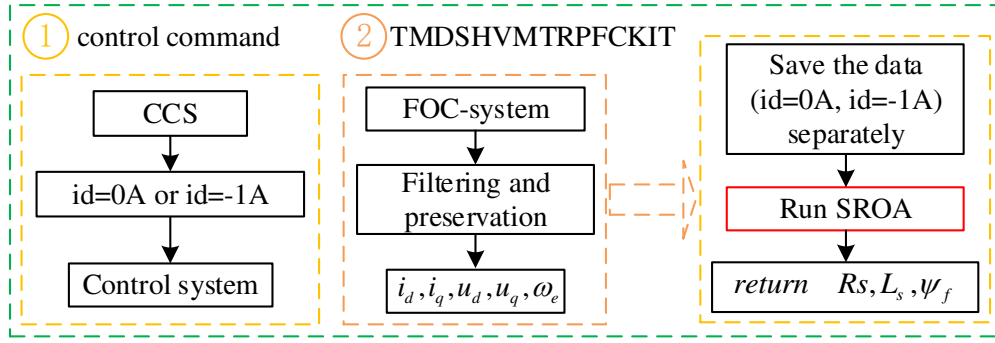


Figure 7. Parameter identification process on a motor control platform.

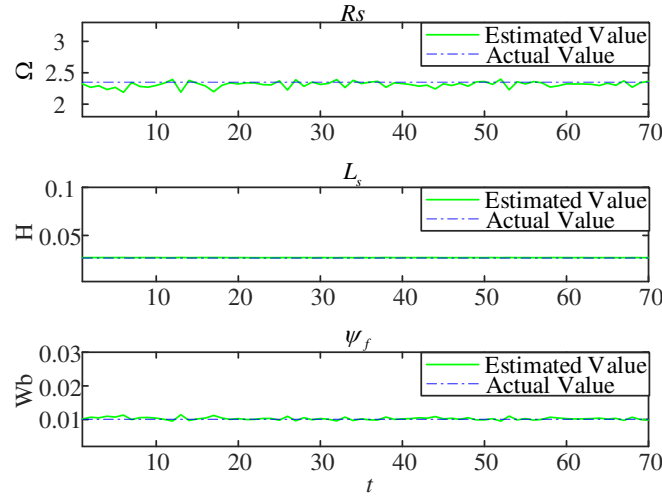


Figure 8. EROA Recognition Results in the Experimental Platform Chip.

Conclusion

In this article, R_s , L_s , and ψ_f of PMSMs were identified based on an enhanced raccoon optimization algorithm. The ROA was improved using the adaptive exploration radius strategy, raccoon-washing-food-inspired strategy, and escaping-predator strategy to form the EROA with better solution efficiency. The simulation results show that the EROA can achieve faster converge speed and higher robustness than the ROA. From the experimental results, the EROA can quickly find the motor parameters, which is consistent with the simulation, and the error of the identification results does not exceed 2%.

Data availability

All data generated or analysed during this study are included in this published article [and its supplementary information files].

Table 6. Statistical analysis of the EROA identification results in the experimental platform.

EROA	$R_s(\Omega)$	$L_s(H)$	$\psi_f(Wb)$
actual value	2.35	0.0265	0.0101
maximum	2.3986	0.0271	0.0114
minimum	2.1886	0.0268	0.0095
average	2.3136	0.0269	0.0103
average error/%	-1.5480	1.6241	1.7775

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Acknowledgements

The authors would like to thank the National Natural Science Foundation of China General Program, Henan Province Key R&D Project, and Hunan Provincial Natural Science Foundation for the scholarships with key codes 62273313, 241111242300, 2023JJ60178, and 2024JJ8046.

Author contributions statement

Z.H. and X.Y. conceived of the presented idea design and simulation; J.Z. and Z.F. carried out testing and verification; J.Z. wrote the original draft of this article; Z.L., X.H. and X.Y. reviewed and edited this article. All authors provided critical feedback and helped shape the research, analysis, and manuscript. All authors have read and agreed to the published version of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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